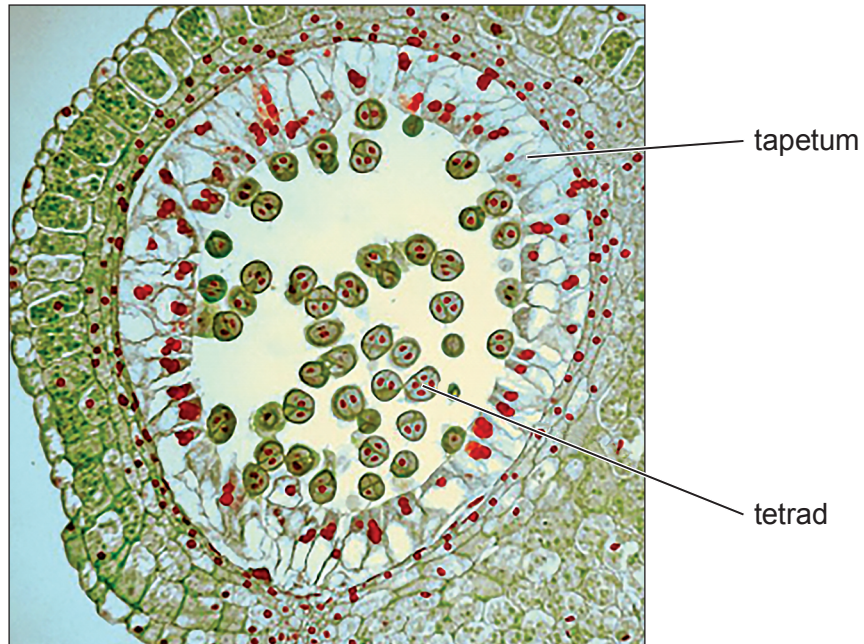


SECTION A

Answer all questions.

1. (a) The photomicrograph below shows a section through part of an anther.



- (i) State the function of the tapetum. [1]

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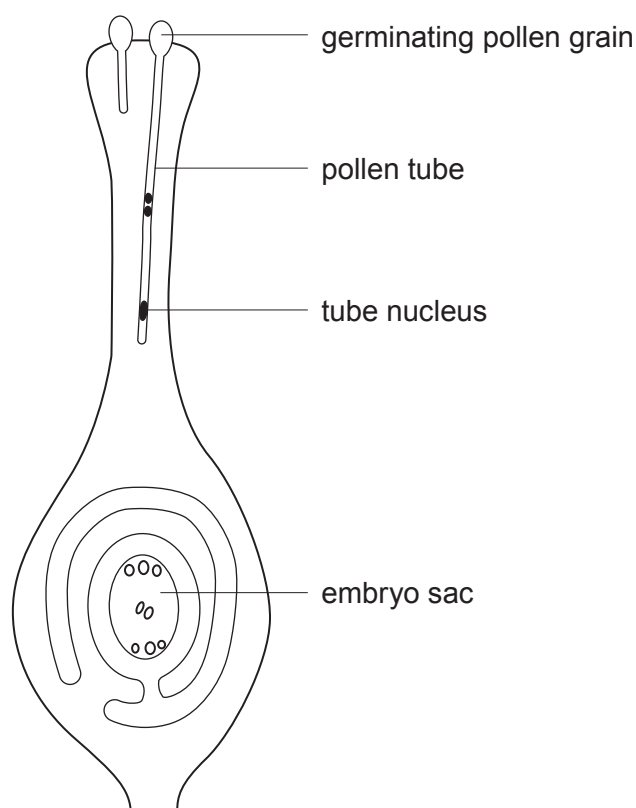
- (ii) Explain the appearance of the tetrad. [2]

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(b) The diagram below shows a carpel following pollination.



(i) Describe the function of the tube nucleus in the growth of the pollen tube. [2]

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(ii) In maize, the distance between the stigma and the micropyle is 20 cm. It takes 20 hours for the pollen tube to grow and reach the micropyle. Calculate the rate of growth of the pollen tube. Give your answer to one decimal place. [3]

Rate of growth = mm minute⁻¹



The photomicrograph below shows an embryo sac in the carpel.



(iii) Use clearly drawn label lines to show the following:

A = Female gamete.
B = Polar nuclei.

[1]

(iv) Describe the role of mitosis and meiosis in the formation of the female gamete. [3]

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(v) With reference to the photomicrograph, explain why fertilisation in plants is referred to as *double fertilisation*. [3]

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